

ST LINA AGCM, AB, Canada

WMO: 715480

Lat: 54.2824N

Lon: 111.4512W

Elev: 2031

StdP: 13.65

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-24.4	-19.3	-30.9	1.0	-24.2	-25.6	1.4	-19.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	81.0	62.9	78.0	62.0	75.2	60.7	66.4	76.5	64.2	74.1	62.4	72.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.3	90.7	72.2	60.2	84.0	69.3	58.4	78.7	66.9	32.2	76.9	30.5	74.0	29.1	72.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-31.5	86.3	5.0	2.2	-35.1	87.9	-38.0	89.1	-40.8	90.4	-44.4	92.0		
(5)			-31.7	70.8	4.9	2.3	-35.3	72.5	-38.2	73.8	-40.9	75.1	-44.5	76.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	11.2	9.9	22.2	37.1	50.7	58.2	62.0	60.0	51.5	38.5	22.2	9.6		
	DBStd	22.32	15.96	15.14	13.78	9.33	8.34	5.82	4.53	5.46	8.11	8.55	12.24	14.35		
	HDD50	6229	1203	1124	863	397	95	11	0	3	78	368	835	1252		
	HDD65	10555	1668	1544	1328	838	447	213	116	168	408	823	1285	1717		
	CDD50	1199	0	0	0	9	116	256	372	314	122	10	0	0		
	CDD65	51	0	0	0	0	3	9	22	15	2	0	0	0		
	CDH74	861	0	0	0	2	99	132	294	236	97	1	0	0		
	CDH80	121	0	0	0	0	14	12	46	30	19	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	19.5	1.0	0.6	0.8	1.2	2.1	3.4	3.6	2.8	2.2	0.7	0.7	0.9		
(15) Precipitation	PrecMax	27.9	2.7	2.4	1.4	2.1	5.0	7.3	6.4	6.6	5.6	2.4	1.8	1.6		
	PrecMin	14.3	0.3	0.0	0.3	0.0	0.6	1.5	1.9	0.2	0.2	0.1	0.0	0.1		
	PrecStd	3.5	0.6	0.5	0.4	0.6	1.2	1.7	1.4	1.6	1.4	0.6	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	44.4	53.3	71.8	82.0	81.7	84.8	83.6	83.2	70.6	53.4	42.2		
		MCWB	37.3	37.4	41.9	50.5	56.8	61.2	66.9	65.0	61.5	54.4	43.2	35.1		
	2%	DB	41.2	39.7	47.0	62.9	76.7	77.8	80.5	79.6	76.4	61.2	45.6	37.8		
		MCWB	35.2	34.9	38.6	45.5	55.5	60.2	65.3	64.0	58.9	49.1	39.0	32.5		
	5%	DB	37.7	35.4	42.7	57.3	71.7	74.5	77.2	76.5	71.6	56.0	41.1	33.3		
		MCWB	33.4	31.3	35.9	43.4	52.9	58.6	63.8	63.3	57.5	45.7	36.5	29.6		
	10%	DB	34.3	32.2	38.8	52.0	67.6	71.2	74.1	73.5	66.2	51.6	37.0	29.2		
		MCWB	30.9	29.4	33.5	40.7	50.7	57.5	62.0	61.2	54.2	43.5	33.6	26.5		
	0.4%	WB	38.1	38.1	42.5	51.4	59.0	64.7	70.7	69.0	62.7	54.8	44.3	36.1		
		MCDB	43.9	43.4	50.9	71.3	74.6	76.3	80.6	78.6	77.1	67.5	51.6	41.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.8	34.8	38.9	46.6	56.7	62.4	67.2	66.3	60.1	50.2	39.9	32.6		
		MCDB	40.3	39.1	46.0	60.8	72.3	73.4	77.1	76.5	73.9	58.2	44.8	37.5		
	5%	WB	33.5	32.0	36.3	44.0	54.4	60.7	65.0	63.9	57.8	46.9	36.8	29.6		
		MCDB	37.5	35.0	42.0	55.5	68.7	71.1	74.6	73.6	70.3	54.1	40.8	33.1		
	10%	WB	30.8	29.3	34.1	41.6	52.1	59.0	63.0	62.0	55.4	44.0	33.5	26.5		
		MCDB	34.4	32.4	38.2	50.7	64.8	68.2	72.1	71.6	64.6	50.9	36.7	29.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	17.7	17.4	18.9	23.7	20.2	20.6	21.6	21.6	16.8	13.6	14.4		
		MCDBR	17.0	19.4	19.8	27.4	29.7	26.0	26.2	27.1	30.9	25.0	17.7	17.2		
		MCWBR	13.3	15.6	13.0	14.4	13.3	12.0	13.6	13.7	15.4	14.8	12.1	13.8		
		MCDBR	16.8	18.1	18.5	25.6	28.3	22.2	23.7	25.4	29.3	23.7	16.8	17.2		
	5% WB	MCWBR	13.5	15.0	12.4	14.3	13.2	11.5	13.3	14.0	15.3	14.5	12.1	14.3		
(40) Clear-Sky Solar Irradiance	taub		0.253	0.264	0.288	0.326	0.344	0.358	0.359	0.365	0.308	0.278	0.251	0.231		
	taud		2.277	2.304	2.355	2.388	2.392	2.367	2.410	2.344	2.515	2.569	2.450	2.300		
	Ebn at Noon		228	267	285	287	286	282	279	270	276	264	234	215		
	Edh at Noon		18	24	30	33	35	36	34	34	25	19	15	14		
(44) All-Sky Solar Radiation	RadAvg		279	609	1093	1478	1809	1852	1886	1536	1079	603	282	196		
	RadStd		30	49	79	124	148	129	102	121	116	96	36	19		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page